



Inspiring Excellence

CSE421 (Computer Networks)

Project: Genome Assembly Research Network

Group: 1501

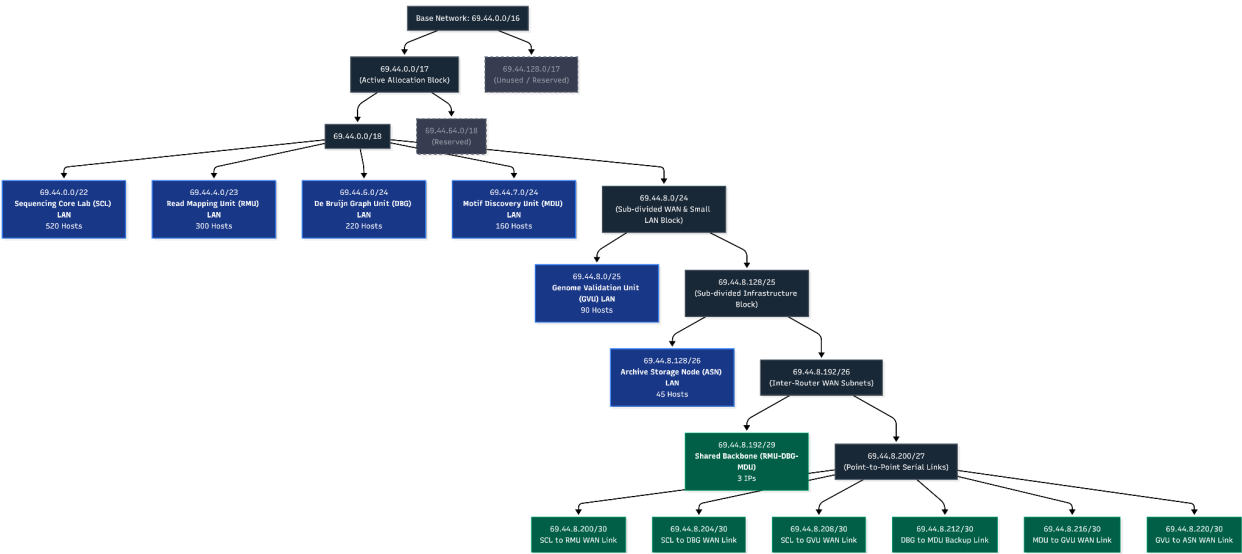
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1. VLSM Tree:



2. VLSM Table:

Network /Link Name	Required Hosts	Network Address	CIDR Prefix	Subnet Mask	Usable IP Range	Broadcast Address
Sequencing Core Lab (SCL) LAN	520	69.44.0.0	/22	255.255.252.0	69.44.0.1 – 69.44.3.254	69.44.3.255
Read Mapping Unit (RMU) LAN	300	69.44.4.0	/23	255.255.254.0	69.44.4.1 – 69.44.5.254	69.44.5.255

De Bruijn Graph Unit (DBG) LAN	220	69.44.6.0	/24	255.255.255.0	69.44.6.1 – 69.44.6.254	69.44.6.255
Motif Discovery Unit (MDU) LAN	160	69.44.7.0	/24	255.255.255.0	69.44.7.1 – 69.44.7.254	69.44.7.255
Genome Validation Unit (GVU) LAN	90	69.44.8.0	/25	255.255.255.1 28	69.44.8.1 – 69.44.8.126	69.44.8.127
Archive Storage Node (ASN) LAN	45	69.44.8.128	/26	255.255.255.1 92	69.44.8.129 – 69.44.8.190	69.44.8.191
Shared Backbone (RMU-DBG-MDU)	3	69.44.8.192	/29	255.255.255.2 48	69.44.8.193 – 69.44.8.198	69.44.8.199
SCL to RMU WAN Link	2	69.44.8.200	/30	255.255.255.2 52	69.44.8.201 – 69.44.8.202	69.44.8.203
SCL to DBG WAN Link	2	69.44.8.204	/30	255.255.255.2 52	69.44.8.205 – 69.44.8.206	69.44.8.207
SCL to GVU WAN Link	2	69.44.8.208	/30	255.255.255.2 52	69.44.8.209 – 69.44.8.210	69.44.8.211
DBG to MDU WAN (Backup)	2	69.44.8.212	/30	255.255.255.2 52	69.44.8.213 – 69.44.8.214	69.44.8.215

MDU to GVV WAN Link	2	69.44.8.216	/30	255.255.255.252	69.44.8.217 – 69.44.8.218	69.44.8.219
GVV to ASN WAN Link	2	69.44.8.220	/30	255.255.255.252	69.44.8.221 – 69.44.8.222	69.44.8.223

3. IP Addressing Table:

Device	Interface	IPv4 Address	Subnet Mask	Default Gateway
SCL Router	GigabitEthernet0/0	69.44.0.1	255.255.252.0	N/A
	Serial0/2/0	69.44.8.201	255.255.255.252	N/A
	Serial0/2/1	69.44.8.205	255.255.255.252	N/A
	Serial0/3/0	69.44.8.209	255.255.255.252	N/A
DNS Server 1	FastEthernet0	69.44.0.2	255.255.252.0	69.44.0.1
Web Server 1	FastEthernet0	69.44.0.3	255.255.252.0	69.44.0.1
Email Server 1	FastEthernet0	69.44.0.4	255.255.252.0	69.44.0.1
DHCP Server 1	FastEthernet0	69.44.0.5	255.255.252.0	69.44.0.1
Printer0	FastEthernet0	69.44.0.6	255.255.252.0	69.44.0.1
PC0	FastEthernet0	DHCP Assigned	DHCP Assigned	DHCP Assigned

PC1	FastEthernet0	DHCP Assigned	DHCP Assigned	DHCP Assigned
RMU Router	GigabitEthernet0/0	69.44.4.1	255.255.254.0	N/A
	GigabitEthernet0/1	69.44.8.193	255.255.255.248	N/A
	Serial0/2/0	69.44.8.202	255.255.255.252	N/A
Printer1	FastEthernet0	69.44.4.2	255.255.254.0	69.44.4.1
PC2	FastEthernet0	DHCP Assigned	DHCP Assigned	DHCP Assigned
PC3	FastEthernet0	DHCP Assigned	DHCP Assigned	DHCP Assigned
DBG Router	GigabitEthernet0/0	69.44.6.1	255.255.255.0	N/A
	GigabitEthernet0/1	69.44.8.194	255.255.255.248	N/A
	Serial0/2/0	69.44.8.206	255.255.255.252	N/A
	Serial0/2/1	69.44.8.217	255.255.255.252	N/A
Printer2	FastEthernet0	69.44.6.2	255.255.255.0	69.44.6.1
PC4	FastEthernet0	DHCP Assigned	DHCP Assigned	DHCP Assigned
PC5	FastEthernet0	DHCP Assigned	DHCP Assigned	DHCP Assigned
MDU Router	GigabitEthernet0/0	69.44.7.1	255.255.255.0	N/A
	GigabitEthernet0/1	69.44.8.195	255.255.255.248	N/A
	Serial0/2/0	69.44.8.214	255.255.255.252	N/A
	Serial0/2/1	69.44.8.218	255.255.255.252	N/A
Email Server 2	FastEthernet0	69.44.7.2	255.255.255.0	69.44.7.1
DHCP Server 2	FastEthernet0	69.44.7.3	255.255.255.0	69.44.7.1
Printer3	FastEthernet0	69.44.7.4	255.255.255.0	69.44.7.1
PC6	FastEthernet0	DHCP Assigned	DHCP Assigned	DHCP Assigned
PC7	FastEthernet0	DHCP Assigned	DHCP Assigned	DHCP Assigned
GVU Router	GigabitEthernet0/0	69.44.8.1	255.255.255.128	N/A
	Serial0/2/0	69.44.8.210	255.255.255.252	N/A
	Serial0/2/1	69.44.8.213	255.255.255.252	N/A
	Serial0/3/0	69.44.8.221	255.255.255.252	N/A
Printer4	FastEthernet0	69.44.8.2	255.255.255.128	69.44.8.1
PC8	FastEthernet0	DHCP Assigned	DHCP Assigned	DHCP Assigned
PC9	FastEthernet0	DHCP Assigned	DHCP Assigned	DHCP Assigned
ASN Router	GigabitEthernet0/0	69.44.8.129	255.255.255.192	N/A
	Serial0/2/0	69.44.8.222	255.255.255.252	N/A
Printer5	FastEthernet0	69.44.8.130	255.255.255.192	69.44.8.129
PC10	FastEthernet0	69.44.8.131	255.255.255.192	69.44.8.129

PC11	FastEthernet0	69.44.8.132	255.255.255.192	69.44.8.129
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4. Router configuration commands for all routers:

Routers	CLI Commands
Sequencing Core Lab Router (SCL)	<pre> hostname SCL no ip domain-lookup ! interface GigabitEthernet0/0 ip address 69.44.0.1 255.255.252.0 ip helper-address 69.44.0.5 no shutdown ! interface Serial0/2/0 ip address 69.44.8.201 255.255.255.252 clock rate 64000 no shutdown ! interface Serial0/2/1 ip address 69.44.8.205 255.255.255.252 clock rate 64000 no shutdown ! interface Serial0/3/0 ip address 69.44.8.209 255.255.255.252 clock rate 64000 no shutdown ! ip route 69.44.8.128 255.255.255.192 69.44.8.210 ! router rip version 2 no auto-summary network 69.0.0.0 passive-interface GigabitEthernet0/0 </pre>

Read Mapping Unit Router (RMU)	<pre> hostname RMU no ip domain-lookup ! ip dhcp excluded-address 69.44.4.1 69.44.4.68 ! ip dhcp pool RMU-Pool network 69.44.4.0 255.255.254.0 default-router 69.44.4.1 dns-server 69.44.0.2 ! interface GigabitEthernet0/0 ip address 69.44.4.1 255.255.254.0 no shutdown ! interface GigabitEthernet0/1 ip address 69.44.8.193 255.255.255.248 no shutdown ! interface Serial0/2/0 ip address 69.44.8.202 255.255.255.252 no shutdown ! ip route 69.44.7.0 255.255.255.0 69.44.8.195 ! router rip version 2 no auto-summary network 69.0.0.0 passive-interface GigabitEthernet0/0 </pre>
De Bruijn Graph Unit Router (DBG)	<pre> hostname DBG no ip domain-lookup ! ip dhcp excluded-address 69.44.6.1 69.44.6.43 ! ip dhcp pool DBG-Pool network 69.44.6.0 255.255.255.0 default-router 69.44.6.1 dns-server 69.44.0.2 ! </pre>

	<pre> interface GigabitEthernet0/0 ip address 69.44.6.1 255.255.255.0 no shutdown ! interface GigabitEthernet0/1 ip address 69.44.8.194 255.255.255.248 no shutdown ! interface Serial0/2/0 ip address 69.44.8.206 255.255.255.252 no shutdown ! interface Serial0/2/1 ip address 69.44.8.217 255.255.255.252 clock rate 64000 no shutdown ! ip route 69.44.0.0 255.255.252.0 69.44.8.218 123 ! router rip version 2 no auto-summary network 69.0.0.0 passive-interface GigabitEthernet0/0 </pre>
Motif Discovery Unit Router (MDU)	<pre> hostname MDU no ip domain-lookup ! interface GigabitEthernet0/0 ip address 69.44.7.1 255.255.255.0 no shutdown ! interface GigabitEthernet0/1 ip address 69.44.8.195 255.255.255.248 no shutdown ! interface Serial0/2/0 ip address 69.44.8.214 255.255.255.252 no shutdown ! interface Serial0/2/1 ip address 69.44.8.218 255.255.255.252 </pre>

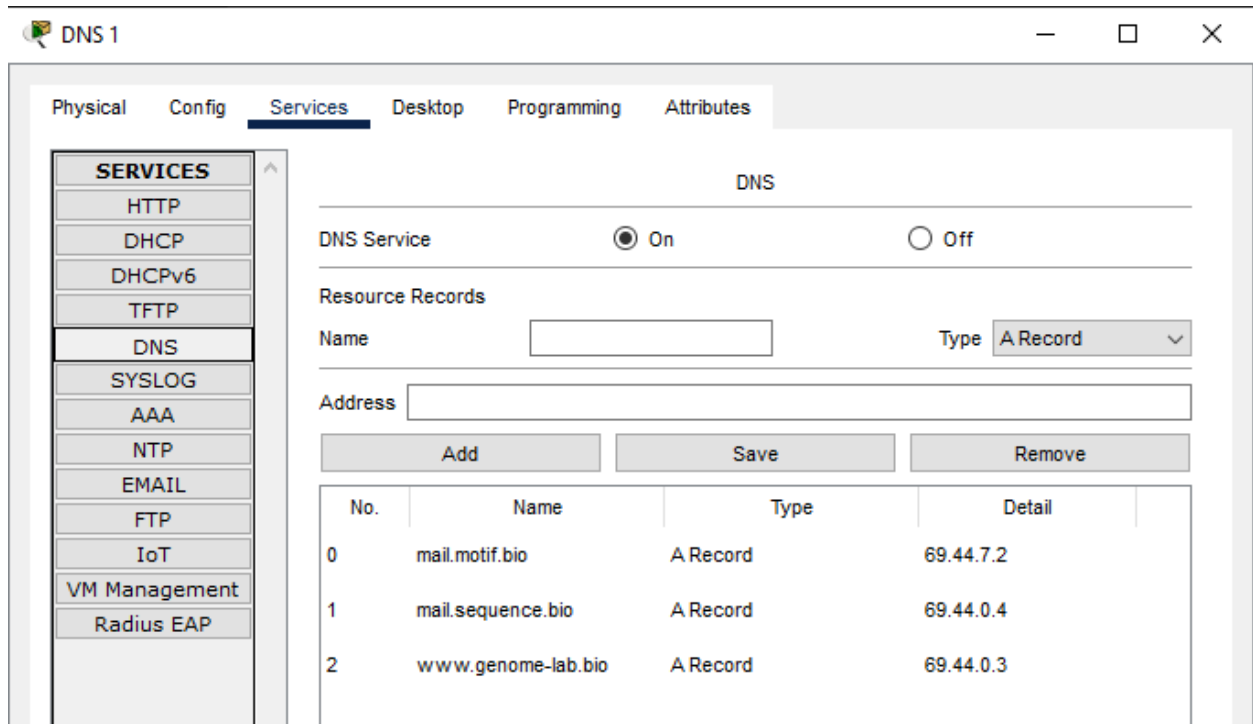
	<pre>no shutdown ! router rip version 2 no auto-summary network 69.0.0.0 passive-interface GigabitEthernet0/0</pre>
Genome Validation Unit Router (GVU)	<pre>hostname Gvu no ip domain-lookup ! interface GigabitEthernet0/0 ip address 69.44.8.1 255.255.255.128 ip helper-address 69.44.0.5 no shutdown ! interface Serial0/2/0 ip address 69.44.8.210 255.255.255.252 no shutdown ! interface Serial0/2/1 ip address 69.44.8.213 255.255.255.252 clock rate 64000 no shutdown ! interface Serial0/3/0 ip address 69.44.8.221 255.255.255.252 clock rate 64000 no shutdown ! ip route 69.44.8.128 255.255.255.192 69.44.8.222 ! router rip version 2 no auto-summary network 69.0.0.0 passive-interface GigabitEthernet0/0 passive-interface Serial0/3/0 redistribute static</pre>

Archive Storage Node Router (ASN)

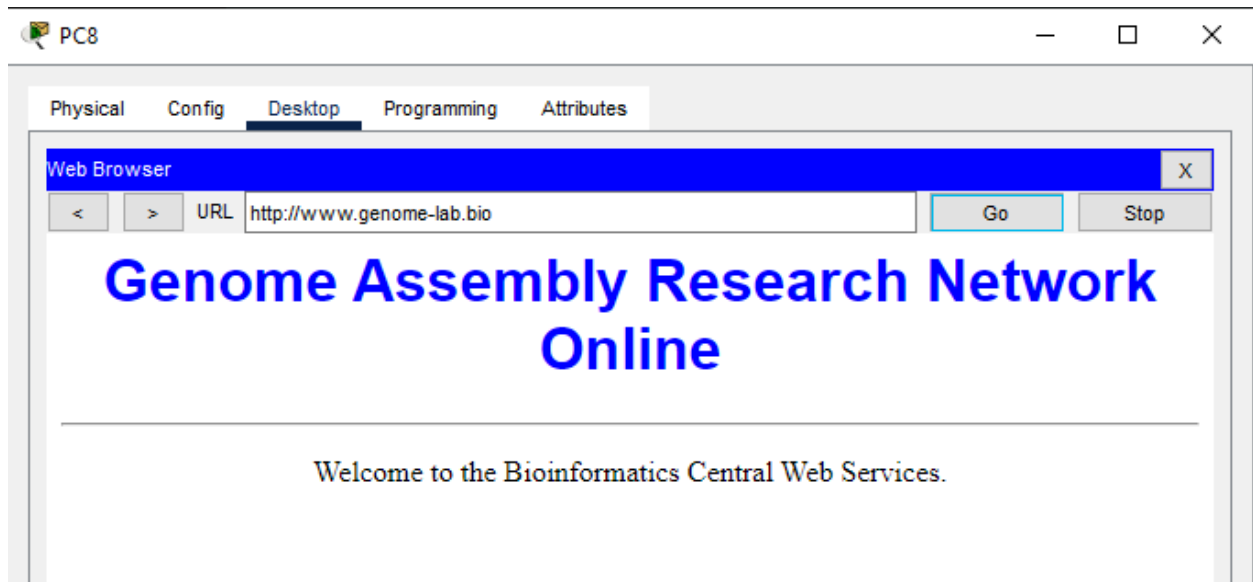
```
hostname ASN
no ip domain-lookup
!
interface GigabitEthernet0/0
 ip address 69.44.8.129 255.255.255.192
 no shutdown
!
interface Serial0/2/0
 ip address 69.44.8.222 255.255.255.252
 no shutdown
!
ip route 0.0.0.0 0.0.0.0 Serial0/2/0
```

5. Server configuration screenshots:

A] DNS:



B] Web:



C] DHCP:

PC2:

The screenshot shows a configuration window for PC2 with tabs for Physical, Config, Desktop, Programming, and Attributes. The Desktop tab is active, displaying the IP Configuration section for the FastEthernet0 interface. The DHCP option is selected, and the configuration fields are populated with the following values:

Field	Value
IPv4 Address	69.44.4.70
Subnet Mask	255.255.254.0
Default Gateway	69.44.4.1
DNS Server	69.44.0.2

The IPv6 Configuration section shows the Static option selected, with the Link Local Address set to FE80::202:17FF:FE0C:C421. The 802.1X section is currently disabled.

Interface: FastEthernet0

IP Configuration

☒ DHCP ☐ Static DHCP request successful.

IPv4 Address: 69.44.4.70

Subnet Mask: 255.255.254.0

Default Gateway: 69.44.4.1

DNS Server: 69.44.0.2

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address: /

Link Local Address: FE80::202:17FF:FE0C:C421

Default Gateway:

DNS Server:

802.1X

☐ Use 802.1X Security

Authentication: MD5

Username:

Password:

PC8:

PC8

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface: FastEthernet0

IP Configuration

☒ DHCP ☐ Static

IPv4 Address: 69.44.8.3

Subnet Mask: 255.255.255.128

Default Gateway: 69.44.8.1

DNS Server: 69.44.0.2

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address: /

Link Local Address: FE80::2E0:B0FF:FE69:EC43

Default Gateway:

DNS Server:

802.1X

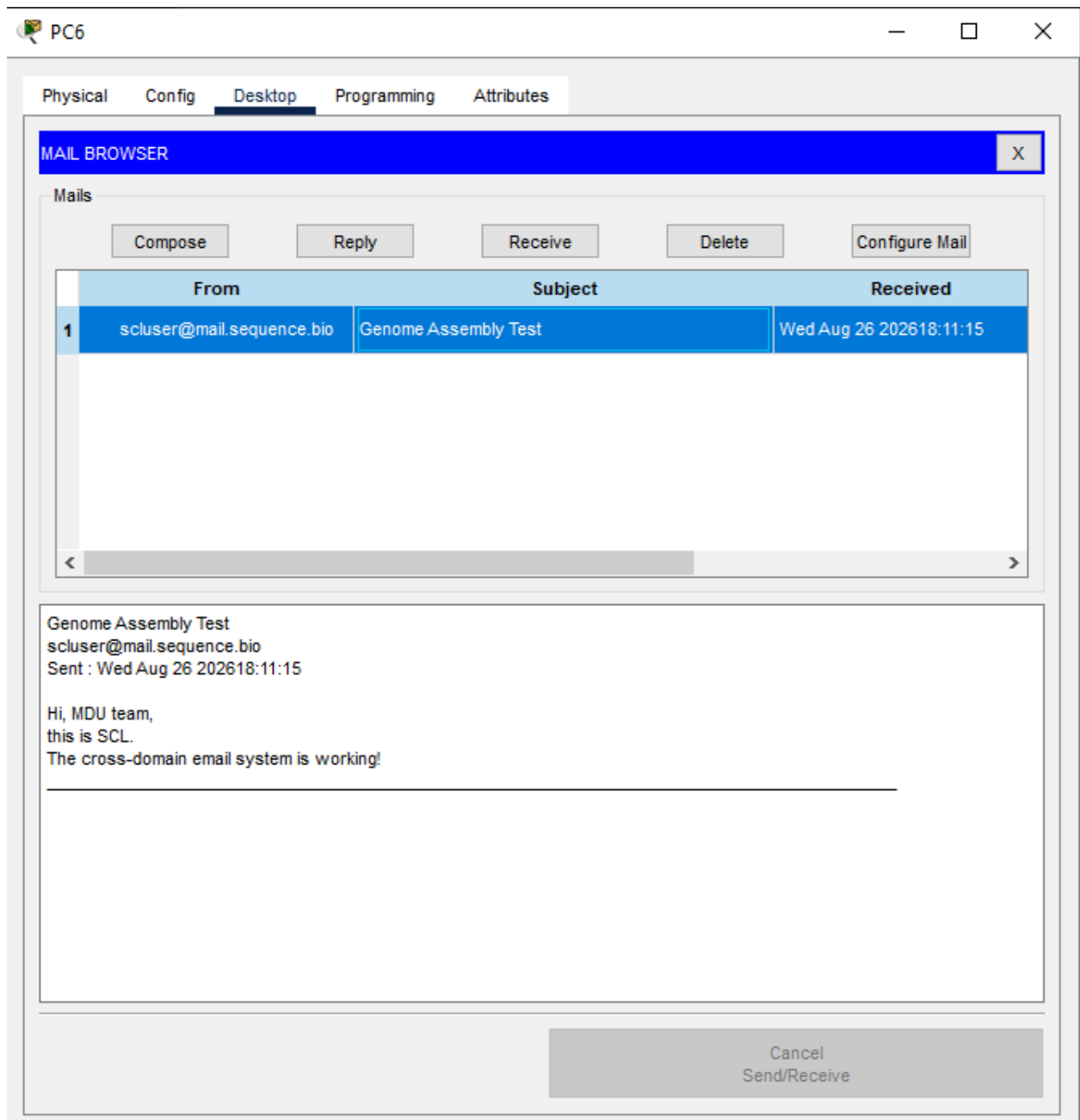
☐ Use 802.1X Security

Authentication: MD5

Username:

Password:

D] Email:



6. Work Distribution Table:

Student ID	Student Name	Assigned Work
22101569	MD. Aminul Islam Ornob	Worked on the Packet Tracer (.pkt) file, network topology, router and switch configuration, server setup, and connectivity testing.
22101844	Afaf Binta Shamsuddin	Completed Deliverables 2–6 including VLSM subnetting, IP addressing table, topology planning, and documentation.
24301043	Dipannita Banik	Assisted with Deliverables 2–6 by preparing subnet allocation, IP addressing, and documentation.
22201312	Syed Isfar Rahman	Prepared Deliverables 7 and 8, including work distribution and explanation of DHCP, static routing, dynamic routing, and network design.

7. Short Explanation:

A. Student ID to Base Network Conversion

The project uses the base network 69.44.0.0/16. Variable Length Subnet Masking (VLSM) divides this network into subnets according to the host requirements of each research unit. Larger LANs receive larger subnet masks, while WAN point-to-point links use /30 subnetworks.

B. DHCP Design

The network uses both server-based DHCP and router-based DHCP. Sequencing Core Lab (SCL) uses DHCP Server1 (69.44.0.5). Motif Discovery Unit (MDU) uses DHCP Server2 (69.44.7.3). Read Mapping Unit (RMU) and De Bruijn Graph Unit (DBG) use router-based DHCP. Genome Validation Unit (GVU) receives IP addresses through DHCP Relay using the ip helper-address command, forwarding requests to the SCL DHCP server.

C. Static Routing Design

Static routing is configured for the Archive Storage Node (ASN), which is outside the RIP routing domain. The ASN network uses subnet 69.44.8.128/26. A floating static route is configured between DBG and SCL through the DBG–MDU backup WAN link. This route becomes active if the primary route fails.

D. Dynamic Routing (RIPv2)

RIPv2 is enabled on the SCL, RMU, DBG, MDU, and GUV routers. These routers advertise their connected LAN and WAN networks so routing information is exchanged automatically. RIPv2 supports classless routing and VLSM, making it suitable for this subnetted network.

E. Router Using Both Static and Dynamic Routing

The Genome Validation Unit (GVU) Router uses both routing methods. It participates in RIPv2 to communicate with other research units while also maintaining a static route to reach the Archive Storage Node (ASN).